Supportive Token Revealing for Fast Diffusion Language Model Decoding

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SCAN FOR CODE

The problem

Diffusion LLMs (LLaDA, Dream) denoise all masked positions **in parallel** - a natural source of speed that autoregressive decoding lacks. But parallelism creates a **quality-latency trade-off**:

- **Aggressive** decoding commits mutually dependent tokens too early → inconsistent text;
- **Conservative** decoding needs many denoising steps → slow generation.

Existing training-free decoders (confidence-based, locality-aware, dependency-aware) all answer one question: *which tokens are **safe** to reveal together?* Yet avoiding unsafe commits does not make the rest of the sequence easier - uncertain tokens may depend on *still-masked* tokens, so decoding stalls on a hard "dependency core."

Key idea

Flip the question: instead of only asking which tokens are safe to reveal *now*, ask which reveals would provide the most **useful context** for the tokens that remain.

AXON reveals **anchors** - confident masked tokens that *uncertain* positions attend to strongly. Like axons in neural systems, they transmit context to where it is needed.

- **Training-free** plug-in: layers on top of any parallel decoder;
- Backbone stays **frozen**; uses only signals already computed in the forward pass (attention maps, confidences);
- Intervenes **only when needed**, via a lightweight gate.

Anchors break the bottleneck

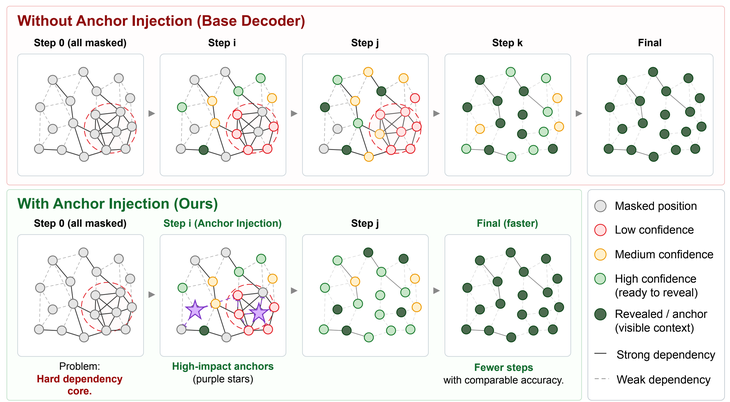


Fig. 2. Purple stars mark anchors that provide context for unresolved dependencies and accelerate later denoising.

Takeaways

- Fast dLLM decoding should go beyond *safe* commit rules and actively **construct useful context**;
- A handful of well-chosen anchors delivers the support of many base commits - submodular coverage keeps them non-redundant;
- Gated intervention makes AXON nearly free when the base decoder is already doing well.

KEY POINTS

- **Training-free** plug-in; the backbone stays frozen.
- **Gated intervention** using existing attention and confidence signals.

How AXON works

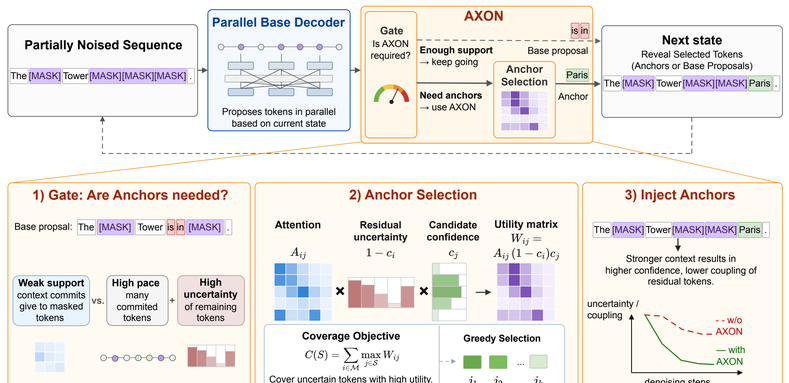


Fig. 1. Gate → select anchors → reveal supportive context.

Algorithm

Algorithm 1 AXON decoding step at time t.

Input: Masked $M^{(t)}$, Uncertain $U^{(t)}$, Base $S^{(t)}_{\text{base}}$, Confidences $c^{(t)}$, Attention $A^{(t)}$, Block $B^{(t)}$, Gate weight α, Budget T, Pace at step t $r^{(t)}$, Coverage objective $C^{(t)}$

Output: Committed token set $S^{\star(t)}$

```
# compute decoding paces
r^(t) ← |S^(t)_base| / |M^(t)|
ρ^(t) ← 1/(T − t)
# compute residual confidence and contextual support
c̄^(t)_M ← (1/|U^(t)|) Σ_{i∈U^(t)} c^(t)_i
g^(t) ← (1/|U^(t)|) Σ_{i∈U^(t)} (Σ_{j∈S^(t)_base} A^(t)_ij) / (Σ_{j∈B^(t)} A^(t)_ij + ε)   (Eq. 4)
# calculate deficits
d^(t)_pace ← (ρ^(t) − r^(t)) 𝕀[r^(t) ≤ r^(t−1)]   (Eq. 3)
d^(t)_ctx ← max(r^(t), 1 − c̄^(t)_M) − g^(t)   (Eq. 5)
# evaluate unified gate (Eq. 6)
if α (d^(t)_ctx)_+ + (1 − α) (d^(t)_pace)_+ > 0 then
    # gate open: dynamic submodular selection
    τ^(t) ← C^(t)(S^(t)_base)   (Eq. 2)
    S ← ∅
    while C^(t)(S) < τ^(t) and M^(t) \ S ≠ ∅ do
        # greedily add anchor with max marginal gain
        Δ_j ← C^(t)(S ∪ {j}) − C^(t)(S)
        j* ← arg max_{j∈M^(t)\S} Δ_j
        S ← S ∪ {j*}
    end while
    S^⋆(t) ← S
else
    # gate closed: commit base proposal
    S^⋆(t) ← S^(t)_base
end if
return S^⋆(t)
```

Original algorithm from the paper.

Submodular selection

1. Score each candidate

$$w_{ij} = A_{ij}\,(1 - c_i)\,c_j$$

Attention × uncertainty × candidate confidence

2. Cover uncertain tokens

$$C(S) = \sum_{i \in U} \max_{j \in S} w_{ij}$$

- Credit each token to its **best selected anchor**.
- **Diminishing returns** discourages redundant anchors.

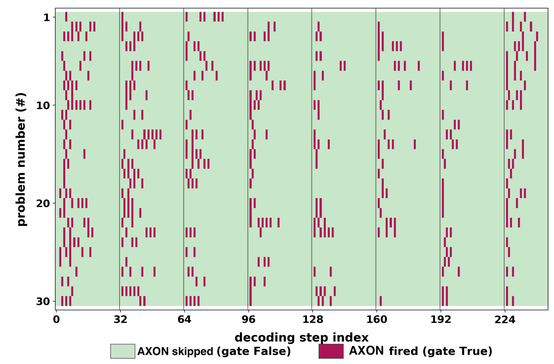
Monotone submodular. Greedy guarantee: **(1 - 1/e)** for fixed-budget coverage.

FEWER EVALUATIONS · DAWN + AXON (k = 1) / HUMANEVAL

18.6%
LLaDA-1.5
NFE 77.3 → 62.9
Accuracy 42.07 → 42.68%

14.8%
Dream-v0-Instruct-7B
NFE 76.2 → 64.9
Accuracy 54.88 → 55.60%

When does the gate fire?



LLaDA-1.5 / GSM8K. Red: AXON fires; green: gate skipped.
Rows: 30 problems. Columns: decoding steps, grouped into 8 blocks.

The gate monitors:

- **Context deficit:** too little support for uncertain tokens.
- **Pace deficit:** behind schedule without acceleration.

Results Table

Method	HumanEval Acc (↑)	HumanEval TPS(↑)	HumanEval NFE(↓)	MBPP Acc (↑)	MBPP TPS(↑)	MBPP NFE(↓)	GSM8K Acc (↑)	GSM8K TPS(↑)	GSM8K NFE(↓)	Minerva-Math Acc (↑)	Minerva-Math TPS(↑)	Minerva-Math NFE(↓)
LLaDA-1.5												
Original	43.90	6.04	256.0	38.80	2.39	256.0	80.36	6.70	256.0	33.36	8.35	256.0
Confidence	43.29	15.61	100.6	38.00	14.11	42.5	80.44	22.95	74.8	**33.38**	22.31	95.4
+ AXON	43.90	18.53	93.6	**39.20**	14.39	42.2	**81.50**	22.57	72.1	33.24	22.61	92.7
+ AXON$_{CVR}$	43.29	19.74	91.0	38.20	14.52	42.1	81.27	23.30	70.3	33.02	23.06	91.1
LocalLeap	41.46	19.83	75.2	38.80	18.50	32.5	80.06	29.46	57.3	32.40	27.64	75.7
+ AXON	**44.51**	20.42	77.6	38.20	16.77	35.3	80.74	26.74	60.2	32.88	25.86	79.7
+ AXON$_{CVR}$	**44.51**	22.91	73.0	37.60	17.13	34.6	81.12	27.64	58.1	32.34	26.47	78.1
DAWN	42.07	22.96	77.3	37.80	19.37	30.7	80.82	30.55	54.2	31.80	28.63	72.9
+ AXON	42.68	26.96	62.9	37.20	20.97	29.3	81.20	33.02	49.5	32.00	30.64	68.0
+ AXON$_{CVR}$	39.02	**29.11**	**59.6**	37.20	**21.49**	**28.3**	80.52	**34.41**	**47.6**	31.30	**31.60**	**66.0**
LLaDA-8B-Instruct												
Original	40.24	18.09	256.0	29.40	6.48	256.0	77.94	7.14	256.0	**33.20**	9.51	256.0
Confidence	40.85	58.96	77.9	29.60	23.69	69.8	78.39	23.48	77.7	32.98	25.14	96.9
+ AXON	40.55	65.45	74.6	32.00	23.51	62.2	**79.30**	23.39	74.7	32.76	25.75	92.7
+ AXON$_{CVR}$	40.55	66.23	73.6	31.40	23.69	61.6	79.15	24.16	72.2	32.30	26.23	91.0
LocalLeap	39.63	71.20	61.4	29.40	29.84	55.0	77.33	30.55	58.9	32.42	31.61	75.6
+ AXON	**42.68**	76.66	63.1	31.40	28.05	51.5	77.48	27.97	62.1	32.62	29.75	79.1
+ AXON$_{CVR}$	42.06	79.39	60.2	30.20	28.61	50.6	75.89	28.90	59.9	32.16	30.52	77.2
DAWN	40.24	80.26	63.2	30.80	31.19	50.0	77.94	31.95	55.3	32.36	32.83	72.5
+ AXON	39.98	92.00	53.2	**32.40**	31.69	44.4	77.41	34.33	50.6	31.84	35.36	67.0
+ AXON$_{CVR}$	39.58	**92.22**	**52.4**	30.60	**32.35**	**42.4**	76.57	**35.89**	**48.2**	30.84	**36.57**	**64.7**
Dream-v0-Instruct-7B												
Original	53.66	16.62	256.0	54.20	5.42	256.0	**76.35**	4.85	256.0	37.80	11.50	256.0
Confidence	**57.32**	38.30	90.4	55.20	26.38	62.7	75.20	19.23	62.3	38.24	24.66	114.8
+ AXON	57.30	41.68	75.4	55.40	25.19	56.0	76.12	18.53	61.6	**38.50**	24.04	112.1
+ AXON$_{CVR}$	57.30	42.90	70.2	55.20	24.98	55.1	75.82	18.44	61.2	37.74	24.20	110.0
LocalLeap	54.27	42.79	76.1	54.60	28.69	44.7	73.24	22.05	54.2	38.10	28.22	97.7
+ AXON	54.90	47.60	65.1	55.60	28.47	41.4	75.06	21.97	51.5	37.90	28.38	91.2
+ AXON$_{CVR}$	54.27	47.80	70.0	55.00	28.25	**40.6**	74.91	22.04	**51.1**	37.42	28.32	**89.5**
DAWN	54.88	44.77	76.2	55.40	29.76	42.8	73.16	22.20	52.08	38.22	28.65	94.5
+ AXON	55.60	49.70	**64.9**	**55.80**	**30.09**	42.6	74.83	22.59	51.3	38.28	**28.89**	93.0
+ AXON$_{CVR}$	54.80	**50.10**	68.8	55.00	29.89	41.8	74.53	**22.67**	51.7	37.76	28.86	91.2

3 backbones × 4 benchmarks × 3 base decoders. Acc (%); TPS = tokens/s; NFE = forward passes. Green: gain over parent. Bold: best per backbone and dataset metric.

github.com/Jerryaa98/AXON · jerryabuayob@gmail.com